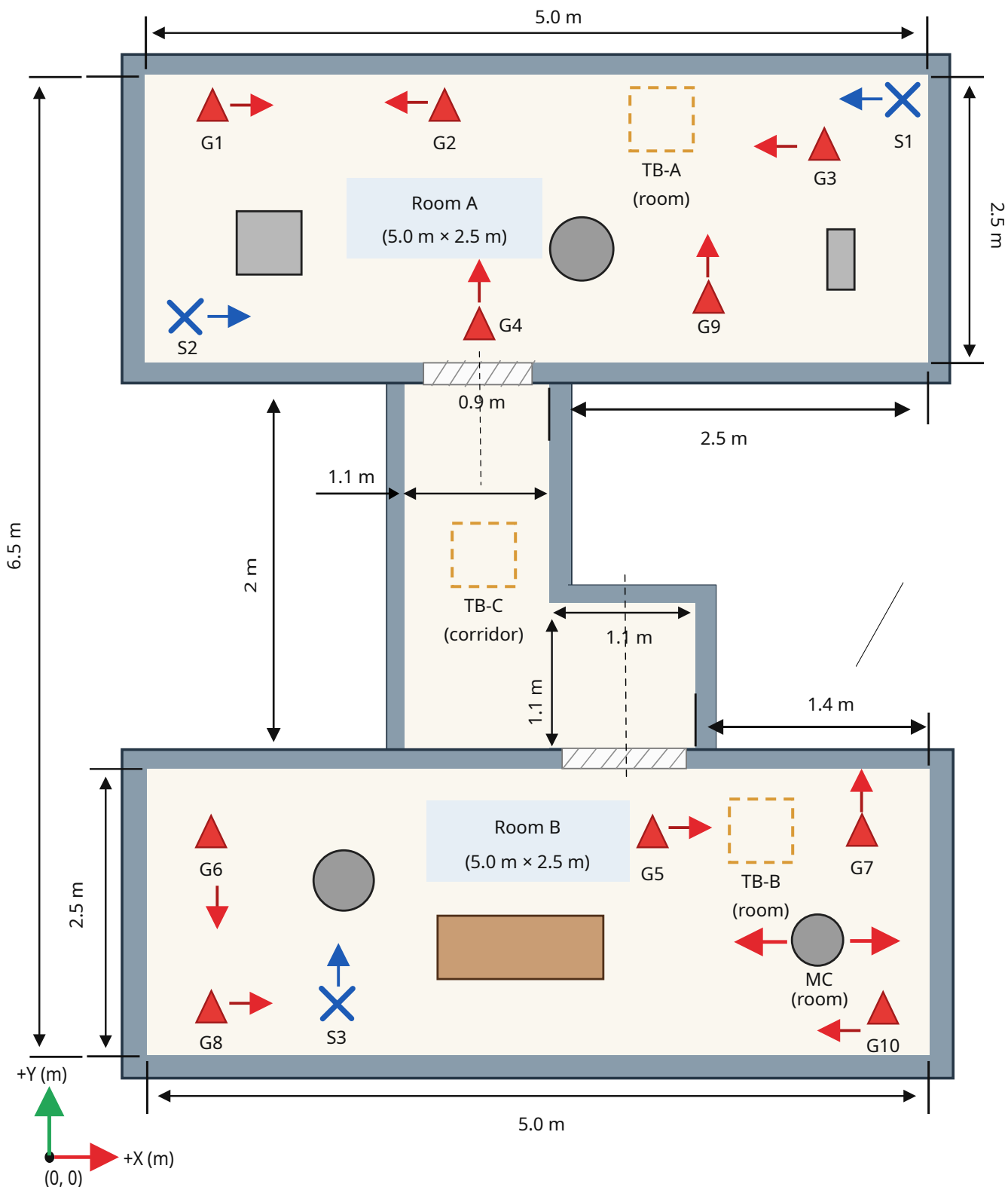


Indoor Navigation Test Environment



Legend

- Start pose (S1, S2, S3)
- Goal pose (G1 – G10)
- Static box (0.50 × 0.50 m)
- Static cylinder (0.45 m)
- Desk / table (1.20 × 0.60 m)
- Localization pillar (0.30 × 0.60 m)
- Temporary box location (0.50 × 0.50 m) (absent during mapping)
- Moving cylinder path (Ø 0.40 m) (stretch scenario, Room B only)
- Doorway (0.9 m clear width)
- Wall (0.2m width)

Start poses (x, y, yaw)

ID	x (m)	y (m)	yaw (°)
S1	4.50	7.55	180
S2	0.45	6.20	0
S3	1.55	1.75	90

Dimensions

Overall size	8.0 × 6.5 m
Room A	5.0 × 2.5 m
Room B	5.0 × 2.5 m
Corridor width	1.1 m (clear)
Doorway width	0.9 m (clear)
Corridor turn pocket	1.1 × 1.1 m

Objects (mapping)

Static boxes	2 (yes)
Static cylinders	2 (yes)
Desk / table	1 (yes)
Localization pillar	1 (yes)
Temporary box	1 (no)
Moving cylinder	1 (no, stretch only)

Goal poses (x, y, yaw)

ID	x (m)	y (m)	yaw (°)
G1	0.55	7.55	0
G2	1.90	7.55	180
G3	4.40	7.50	180
G4	1.70	6.00	90
G5	3.20	1.85	0
G6	0.40	1.00	270
G7	4.55	1.80	90
G8	0.50	0.45	0
G9	3.20	6.20	90
G10	4.50	0.45	180

Temporary box scenarios

ID	Location	Purpose
TB-A	RoomA	Obstacle removal and costmap clearing test
TB-B	Room B	Obstacle removal and costmap clearing test
TB-C	Corridor	Block corridor (goal unreachable)

Moving cylinder scenario

ID	Location	Motion	Purpose
MC	Room B	Horizontal (left ↔ right)	Dynamic obstacle crossing (stop / avoid and continue)